

Football Match Result Prediction Using Neural Networks and Deep Learning

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Abstract—In the present world, the prediction of the results of football matches is being done by both football experts and machines. Football as a game produces a huge amount of statistical data about the players of the team, the matches played between the teams, the environment in which the match is being played. This statistical data can be exploited using various machine learning techniques to predict various information related to a particular football match namely the result of a particular game, injury of a player, performance of a player in a particular match, spotting new talents in the game etc. We in this project will attempt to design a prediction system powered by machine Recurrent Neural Networks and LSTMs.

Keywords—Neural Networks, LSTMs, Recurrent Neural Networks.

I. INTRODUCTION

Football being one of the most popular sports around the globe has a huge amount of fan following the day to day events in the game. The people associated with the football teams as well as the followers of the teams often come across instances where it is hard to guess how their team will perform in a particular game, this is where game result prediction systems come into play. Soccer score prediction can be a helpful means to assess the readiness of a team before going into a game. The prediction can also help the management of different football associations in getting their teams ready for the upcoming matches so as to get the best result possible out of the fixture. Accurate football match outcome prediction is valuable for small and big sports telecommunication companies as it can help them increase their revenue and make the game a lot more interesting for the viewers.

There have been several attempts at trying to predict the outcome of a football match but none of them has been able to match the accuracy of the human prediction. Humans are still by far superior in predicting the outcome of the match as they take into account the technicality as well as the emotional factors that affect the outcome of the game, this helps them in predicting well but it has its drawbacks as well. Humans let their emotions overpower them in the prediction which more or less leads to a wrong prediction of the outcome. Various factors affect the outcome of a football match such as the number of goals scored in previous matches by a team, the winning streak that the team is coming to play with in the present game, the environment in which the team is playing the current fixture, the number of cards that the players involved in the game are playing with and many more other factors associated with the current form of the players in the team as well as the current form of

the team itself. The aim of this paper is to predict the result of a football match. For the prediction process, we make use of Recurrent Neural Networks(RNNs) and Long Short Term Memory(LSTM). We aim at increasing the accuracy of the prediction by taking into consideration all the events that take place during a football match and their effects on the outcome of the match.

II. PROPOSED METHOD

Previous works on predicting the results of football matches with machine learning techniques have mainly focused on the data available about the team. The focus has only been limited to a small number of leagues as well. The approach of this project is to deduce better features from the results of the previous matches that the team has played and taken into consideration the current form of the team predict the accurate result to the game.

A match is played in a particular environment and the events happening on the field have a huge effect on the outcome of the match. So, a Long Short Term Memory system is exploited for this paper.

A. Dataset

The dataset for the project has been taken from "<http://football-data.co.uk/data.php>". The dataset contains the data of the seasons from 2010-11 to 2017-18 of the English Premier League. The advantage associated with this dataset is that the number of matches played by each team and the total number of matches in the tournament is fixed. This was beneficial in removing unnecessary information from the system. In each dataset, there are around 380 records each which are in chronological order with around 60 attributes. From the given dataset manual feature generation was done to get new attributes. For example: the values from the 'FTAG'(Full Time Away Goals) and 'FTHG'(Full Time Home Goals), two new attributes 'HTGS'(Home Team Goals Scored) and 'ATGS'(Away Team Goals Scored) were calculated by summing over the attributes through all the rows. All the contenders in every match had their form calculated by finding the winning streaks of the teams and recording if any team had won 5 games in a row or 3 games in a row. Finally, one hot encoding was done for classification purpose.

B. Methodology

There are a few parameters in every neural network that won't change throughout the working of the model even when the calculation of the accuracy of the model is done. Such parameters are known as hyper-parameters. Once the model has made use of a certain set of hyper-parameters

these parameters can be played around with to get a better understanding and accuracy of the model.

The equation for the value of a state of an RNN cell is given by

$$h_t = f(h_{t-1}, X_t)$$

where h_t is the new value of the state, h_{t-1} is the previous value of the state and X_t is the input. RNNs can not only remember the sequence in one instance, but also the sequence of the various instances coming in because the nodes storing the weights and the activation functions are the same. With respect to football match outcome prediction, the dataset is a relational database as mentioned in section 4.1. Also, the order of the various columns is unimportant as it doesn't affect the outcome of the match.

LSTM cells are a modification of RNNs. LSTM cells implement RNNs and with that, they perform an additional calculation with previous output from the RNN cell and the new input to generate the new output. LSTM cells are mainly composed of three gates :

- Forget gate: this gate is known as the sigmoid function. It is responsible for forgetting information that is not required anymore by the system when moving to the next sequence.
- Input gate: this gate works as an input provider to the network. It uses tanh function as well as the sigmoid function.
- Output gate: this gate is responsible for giving the output that is calculated by the processing that is done at the previous two gates

The given model is implemented with the help of tensorflow library in python. All the data in this library is processed in the form of an array. A code written in Tensorflow works in two steps. Initially, a computational graph is formed, where all the connections among various layers of the neural network and the calculations of the weights are done but none of these variables has values. They are initialized with Tensorflow objects without any numerical values. The second step is the execution, where a session function is run and all the variables that have to be calculated are passed as parameters. This generates numerical values that are assigned to the variables.

C. Implementation

The code for the implementation of LSTM and RNN for the prediction can be seen below.

Pseudocode :

Step 1: import the necessary libraries

import tensorflow as tf

from tensorflow.contrib import rnn

import pandas as pd

Step 2: import the one hot dataset using pandas

Step 3: Define the hyper-parameters

hm_epochs=10

n_classes = 2

batch_size = 1

chunk_size=27

n_chunks=1

rnn_size=512

Step 4: Create the RNN model with LSTMcell.

Step 5: Feed the system with data and Lstmcell mechanism.

Step 6: Train the model using the function created in step 4.

The values of hyper-parameters can be played around with to check which combination of the parameters is best for the prediction system.

III. RESULTS

The experiment has been performed on the above dataset using a basicLSTM cell from the tensorflow library. The model was run on a various set of hyper-parameters to find out the best model. The hyper-parameters include -

- 1) n classes - total number of output classes.
- 2) batch size - number of rows fed in one iteration to the model.
- 3) hm epochs - number of epochs the model runs for.
- 4) chunk size - number of attributes in each chunk.
- 5) n chunks - number of chunks in one instance of data.
- 6) rnn size - number of LSTM cells in the hidden layer of the recurrent neural network.

Table 1 shows the various train and test accuracies with chunk size = 3 and n chunks = 9, for batch sizes from 1 to 124. The model with chunk size as 27 was considered for further analysis. Table 2 shows the accuracies of the model with varying size of the hidden layer. Here, for increasing rnn size, though the training accuracy is increasing, there is no sufficient increase in the test accuracy. This suggests that the model is over-fitting the data. Hence, for the present dataset, the best model values are as follows -

- 1) n classes - 2
- 2) batch size - 1
- 3) hm epochs - 10
- 4) chunk size - 27
- 5) n chunks - 1
- 6) rnn size - 64

TABLE I. RNN WITH CHUNK SIZE 27 AND VARYING BATCH SIZE

Batch size	1	30	60	124 ↑
Accuracy train	0.9811828	0.7688172	0.7413978 6	0.7354839 ↓
Accuracy test	0.8075	0.69875	0.69375	0.69 ↓

TABLE II. :RNN WITH CHUNK SIZE 3 AND N CHUNKS AS 9 AND VARYING BATCH SIZE

Batch size	1	30	60	124 ↑
Accuracy train	0.922043	0.6973118	0.6655914	0.6704301 ↓
Accuracy test	0.79875	0.6525	0.64625	0.6525 ↓

TABLE III. RNN WITH DIFFERENT SIZES OF HIDDEN LAYER

RNN size	32	64	256	512
Accuracy train	0.9731183	0.97043014	0.9919355	0.9892473
Accuracy test	0.8125	0.805	0.805	0.8025

IV. CONCLUSION

The popularity and international effect that football has makes it an interesting problem to solve. Moreover, the number of factors that affect the outcome of a match is enormous. From the results, we can say that RNNs with LSTM show a visible and obvious advantage over the original ANN and the traditional machine learning. Hence, other than the mainstream uses of LSTMs, this particular path of using it for prediction of the outcome for sporting events has also shown promising results. This model can still be improved. One way to do that is, by using a better set of attributes. They can include statistics of each individual player. This can also help in predicting the form of a particular player from season to season. Larger datasets will also help in better training the neural network.

An LSTM network performs wells on classification and has the potential to perform well on predictions of the outcome of football games. In their research Hucaljuk et. al. showed that the Artificial Neural Network predicted the winners of a football match with the best accuracy of 68.8%. In contrast to this, the accuracy achieved by the prediction system that worked with the LSTM form of RNNs showed huge improvement with a test accuracy of 80.75%.

This is no surprise since more information should lead to better predictions. However, the increased prediction accuracy over minutes played in a match indicates that the network is able to learn about football. Future work can be performed on this subject; for example, different data could be used, different inputs and architectures could be tested and other things than winner could be predicted, such as the number of goals or cards.

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